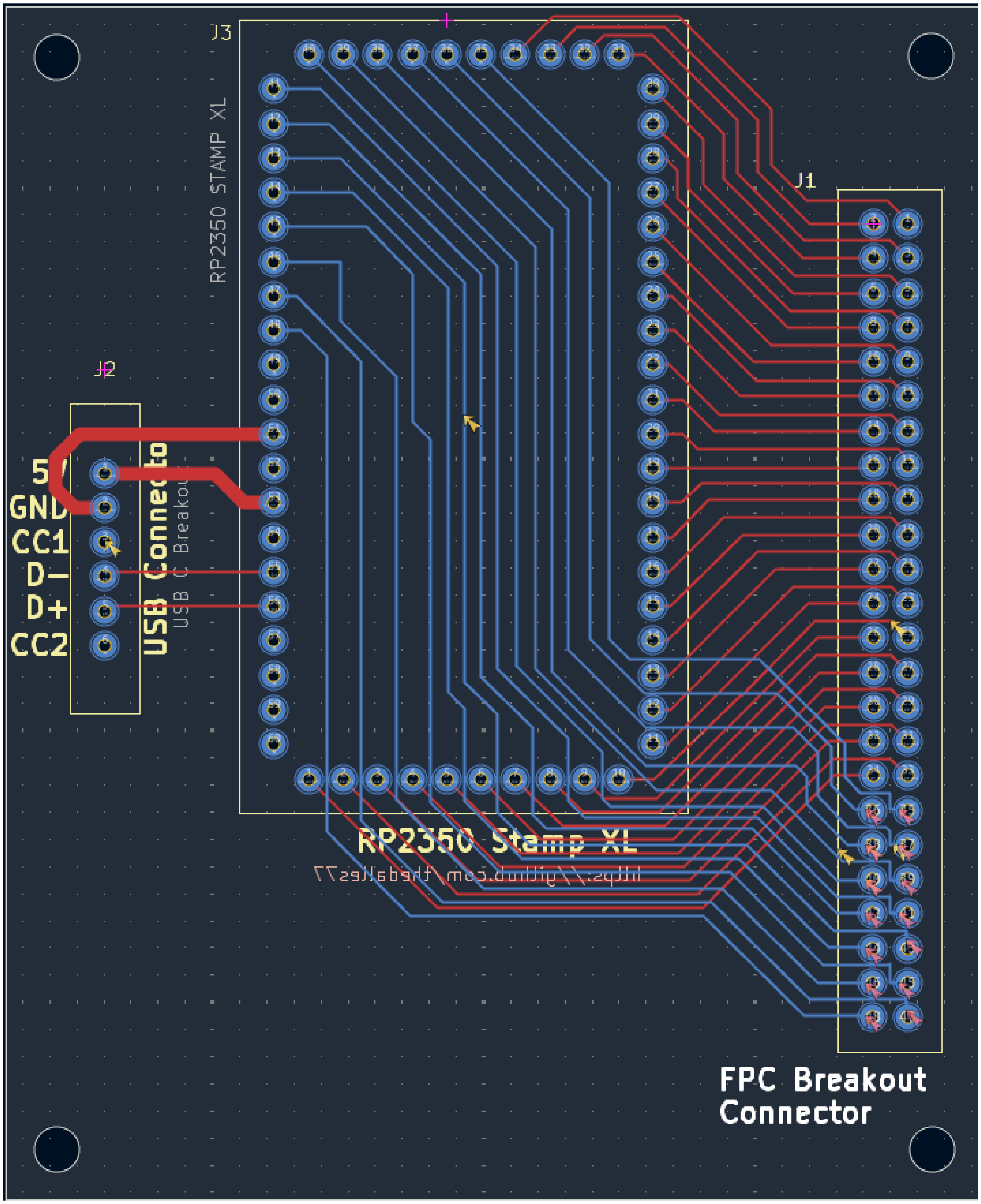
Solder Party RP2350 Stamp XL – 48 Pin Keyboard

The Stamp XL has 48 GPIO's and none are assigned to circuitry on the module. The carrier board shown below routes all 48 GPIO pins to pads for a standard FPC breakout adapter board.



48 pins is not a standard FPC breakout adapter board so purchase a 60 pin board without header pins from AliExpress. If you want it to be able to work with FPC cables with tabs (and you don't want to cut them off), center the 60 pin adapter board on the carrier board and solder it with 48 header pins. The 6 unused connections on each side should be enough space to slide in the tabs. Obviously, you will need to manually align the FPC when you install the cable into the connector. Use a fine tip marking pin to give yourself some alignment guide marks on the connector. Load the Stamp XL with the matrix decoder code so you can push a key and see if it's working.

The first 34 GPIO's (0 - 33) are connected to the same FPC pins as the original RP2350 Stamp XL carrier board. The remaining GPIO's (34 - 47) go to the new FPC pins. New matrix decoder software "Matrix_Decoder_RP2350B_48.py" has been created to add the extra GPIO's.